

# Ice Core

Season Science Lead • Vestri Dome

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15 DAYS • 61 STOPS • GRADE 12 • ICECORE

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## THE OPENING CARD

Vestri Dome is a drilling camp on the polar plateau: six huts up on legs, a trench with a tower over it, and two and a half kilometres of ice already pulled out of the ground. None of it can be let above minus twenty. You are the season science lead, which means the climate record this station sends home is the one you signed. A camp four hundred kilometres away has published a record that disagrees with yours. The aircraft that takes everybody off the ice arrives in fifteen days. Whatever has not been measured by then is not measured until the next crew flies in, two years from now.

## What the plateau has been writing down

### BEFORE THE DAY — GREET

#### **Fourteen people, one season, and the aircraft leaves on Wednesday**

The camp is fourteen people who will be within fifty metres of each other for six weeks, and half of them arrived on the last flight. Halvorsen, the chief scientist, wants as many of them known to you as you can manage before the drill turns, because on a plateau the only thing you cannot fix later is a working relationship.

### PLAN CARD 92W

First morning of the season, and the drill has not turned yet. The field glaciologist, Nadia Brandt, has a fortnight of stake readings and a fresh pit wall, and the chief scientist, Marit Halvorsen, wants the season's accumulation figure before the science plan is written. Today you settle what this site actually records, how much of a year the surface adds, and why the hole was put here rather than 40 kilometres downslope. Get the accumulation wrong and every age below it is wrong in the same direction, quietly, for three kilometres.

### BRIEFING 13W

A new season opens on a core that is already 2000 metres deep.

### WHAT YOU DECIDE 15W

Establish what this site records, how fast it records it, and why anybody drilled here.

## 1.1 How thick a year is here

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · BALLPARK

### SCENE 32W

Brandt has the pit wall photographed and the last summer surface marked at 32 centimetres down. The snow above it weighs a third of what the same thickness of solid ice would.

### GUIDE 68W

Five numbers, and two of them belong elsewhere: the density of liquid water, and the days of stake readings. Ask of each whether the conversion needs it. Snow and ice are the same material at different densities, so a thickness of snow is not a thickness of record. Only ice equivalent lets two sites be compared, and this site gets a tenth of what a coastal one does.

### ASK 7W

Estimate this year's accumulation in ice equivalent.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

snow becomes ice by being compressed under the snow that falls on it ·  
accumulation rate, and what a site with little snow buys and costs — taken as read

### VERDICT 102W

Snow and ice are the same material at different densities, so a thickness of snow is not a thickness of record. Multiplying the layer by the ratio of the two densities converts it to the ice it will become once the air is squeezed out, which is the only form in which two sites can be compared. Vestri gets about 12 centimetres a year in ice equivalent. A coastal site gets a metre, and that difference decides everything downstream: how far back the core reaches, how thin the layers get, and how much older the ice is than its own trapped air.

### TAKEAWAY 15W

An accumulation rate means nothing until it is stated as the ice it will become.

## 1.2 Why the hole is on the dome

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · CHOICE

### SCENE 37W

The drill engineer, Pär Lindqvist, has the hole at 2,040 metres and a map on the wall with 3 abandoned sites on it, all of them on slopes. Vestri Dome is the high point for 200 kilometres.

### GUIDE 69W

Four things a dome might buy, and three of them are about the ice being cold, thick or flat. Ask of each whether it is about where the ice came from. An ice sheet flows outward from its high points. On a slope the ice at a given depth arrived from upstream, so the record mixes one place's weather with a journey. Domes pay for that in thin layers.

### ASK 14W

What does drilling at the dome buy that a site on the slope cannot?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

ice spreads outwards from the high points of an ice sheet · why ice keeps a record at all: burial without melting — taken as read

### VERDICT 88W

An ice sheet flows outwards from its high points. On a slope the ice at a given depth arrived from somewhere upstream, so a core there mixes the weather of one place with the history of a journey, and the two cannot be separated afterwards. Under a dome the flow is almost straight down, so the column below the drill is very nearly the snow that fell on it. Domes pay for that: they are high, cold and dry, which means thin layers and a long slow season.

### TAKEAWAY 18W

A core is a column of what fell above it only where the ice has not travelled sideways.

### 1.3 What makes one year different from the next

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · PROTOCOL

#### SCENE 33W

A metre of core from 60 metres is on the light table. The chronology lead, Ines Okonkwo, has four features marked on it and wants each one attributed before the section is cut.

#### GUIDE 63W

Four features on one metre of core, and four things that could have made them. Pair them by asking what each feature is physically made of: crystals, acid, mineral grains, or bubbles. Then ask what season each belongs to. No single marker survives the whole core, so a year is where several independent signals agree. Where only one does, it is a candidate.

#### ASK 6W

Read a year off the core

#### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

#### TAKES AS READ

snow that falls in summer sits under different conditions from snow that falls in winter

#### VERDICT 86W

No single marker survives the whole core, so a year is established where several independent signals line up. Coarse crystals form on a summer surface that stayed exposed; conductivity spikes are acid, and volcanic acid arrives in a season; dust is delivered by winds thousands of kilometres away and peaks with their season; the light and dark banding is bubble structure changing as each layer was compressed. Where three of the four agree, a year is a year. Where only one does, it is a candidate.

**TAKEAWAY 15W**

A year is counted from several signals that agree, not from one that looks convincing.

**END OF THE DAY 14W**

A record is only as good as the rate at which it was written.

## Three kilometres of one attempt

### BEFORE THE DAY — TRIAL

#### Walk the camp while the weather is holding

Modules on legs, flag lines between them, and a trench under the drill tower. Walk it all once now, in clear air, because the next time you cross the camp in a hurry it will be in blowing snow where the flag lines are the only thing telling you where the next module is.

### PLAN CARD 97W

Second day, and the drill is going back down after a night of maintenance. Lindqvist has three short runs on the sheet from last season's brittle zone and the core line technician, Sigrid Holt, has a section that arrived without its depth mark. Today you decide how the run through the brittle ice is made, what happens to a section that cannot be placed, and what a dirty sample costs. Ice is the only instrument here that gets one reading: what breaks on the way up cannot be drilled again, and what is contaminated cannot be un-contaminated.

### BRIEFING 17W

The brittle zone starts at 540 metres and the run sheet has 3 short runs on it.

### WHAT YOU DECIDE 16W

Get the core to the surface and onto the line without changing what it can say.

## 2.1 The run through the brittle zone

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · SEQUENCE

### SCENE 39W

The barrel is at 780 metres, inside the depth range where the ice comes up cracked. The winch, the fluid and the storage are all set for ordinary ice and Lindqvist wants the order settled before the lift starts.

### GUIDE 64W

All five of these will happen, so ask what each one buys the next. Between about five hundred and twelve hundred metres the bubbles are at pressures the surface cannot support. Ask of each step whether it reduces the shock, lets the pressure equalise, or fixes information that is about to be lost. One of them destroys the depth record if it comes early.

### ASK 6W

Bring a brittle section up whole

### BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

### TAKES AS READ

gas trapped at depth is at the pressure of the ice around it · why ice keeps a record at all: burial without melting — taken as read

### VERDICT 83W

Between about 51200 metres the bubbles are at pressures the surface cannot support, and the ice around them fractures as it is brought up. Each step buys the next one: a slow lift reduces the shock, resting the core at hole temperature lets the pressure equalise gradually, logging the breaks fixes what depth each fragment came from before anything moves, and storing it for weeks lets it finish cracking. Cutting first destroys the depth information and cuts a section that is still moving.

### TAKEAWAY 11W

Ice recovered badly is a shorter record, not a cheaper one.

## 2.2 The metre with no label

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CHOICE

### SCENE 33W

A section has arrived at the line with a broken depth mark. Holt can place it to within about four metres from the run sheet, and the isotope team has asked for it.

### GUIDE 72W

Four options for a section that can only be placed within four metres. Ask of each what the depth is for. It is what turns a measurement into a statement about a year, and four metres here is thirty-odd years. Some work does not use depth at all. And note the real rule: the uncertainty has to travel with the sample, because the danger is somebody downstream who never hears about it.

### ASK 7W

What should the section be used for?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

a measurement belongs to the depth it came from, and to no other

### VERDICT 93W

A depth is what converts a measurement into a statement about a year. Four metres here is thirty-odd years, so an isotope value from this section cannot be placed in the record and would sit in it as a point with no time. It is still perfectly good ice for anything the depth does not enter: density, crystal fabric, how the drill is performing. The rule that matters is that the uncertainty travels with the sample, because the danger is not the gap — it is somebody downstream who never hears about it.

### TAKEAWAY 12W

A sample without a depth is a measurement of nothing in particular.

## 2.3 The sample that was clean until it was handled

COLD LABORATORY · A ROOM · DIAGNOSIS

### SCENE 33W

The cold laboratory technician, Elena Cruz, has a sodium result four times higher than the run before it. She has the blank, the run order and the section's handling record on the bench.

### GUIDE 65W

Five readings, and the quiet ones do the work. The blank before is ordinary and the blank after is high. The trim was done to specification. The replicate core holds the same year with no sodium in it. Ask of each candidate how many of the five it covers. What identifies the source is not the loud number but which of the others stayed quiet.

### ASK 7W

Find out where a signal came from

### BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

### TAKES AS READ

a blank is run through the same process as a sample, with no sample in it · what a proxy is: the chain between the thing and the number — taken as read

### VERDICT 86W

The blank before the sample is ordinary and the blank after it is high, which places the contamination inside the run rather than in the ice. The trim was done to specification, so the outside of the section is ruled out, and the replicate core holds the same year with no sodium in it, so the atmosphere is ruled out as well. Every clean reading here is doing work: what identifies the source is not the loud number but which of the quiet ones stayed quiet.

### TAKEAWAY 15W

A blank measures the process, so anything it shows was never evidence about the ice.

## 2.4 The level that holds the hole open

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · HOLD

### SCENE 43W

The borehole is three kilometres deep and the ice around it is trying to close it. The fluid column is what holds it open, and the level in the tower has to stay where the pressure balance needs it while the drill runs.

### GUIDE 58W

Hold the fluid level inside the band on the tower gauge. The band narrows as the hole deepens, because the deeper the hole the less margin there is between closing it and fracturing it. The pump between the tank and the hole is your control, and every run of the winch changes the balance until you answer it.

### ASK 8W

Hold the fluid level while the drill runs.

### BEHIND THE BUTTON 142W

Why the hole needs holding at all. Ice at three kilometres is under enough pressure to flow, and it closes a borehole slowly and permanently. A column of fluid at close to the ice pressure holds it open, which is the only reason a deep hole can be re-entered at all.

Why too much is as bad as too little. Overpressure fractures the wall and takes fluid into the ice, which is both a lost hole and a contaminated record. The band is two-sided for that reason, and it is not symmetric in consequence.

Why the drill itself moves it. Pulling the drill out displaces less than it did going in, chips come up with fluid on them, and cold fluid coming off the reel is denser than what is in the hole. Each of them is a rate, not a step.

### TAKES AS READ

a column of fluid exerts pressure that grows with its depth · why ice keeps a record at all: burial without melting — taken as read

### VERDICT 174W

Ice this deep flows, and a borehole through it closes on its own unless something holds it open. That something is the fluid column, whose pressure at the bottom has to sit close to the pressure of the ice around it — a little under and the wall creeps in, a little over and it fractures and swallows fluid. Everything in a drilling run disturbs that balance and none of it is a single step. Pulling the drill removes displacement continuously while the winch runs; chips come up carrying fluid; cold fluid off the reel is denser than the column it joins and keeps sinking into it. So the pump is set to a new rate and held rather than tapped. The band narrows with depth because the two failure modes converge: the deeper the hole, the smaller the gap between the pressure that closes it and the pressure that breaks it, and both ends of that gap cost the same thing — a hole nobody can re-enter and a season nobody can repeat.

### TAKEAWAY 15W

A hole is held open by a balance, and a balance is held by somebody.

END OF THE DAY 11W

Everything the core can say is decided before anybody measures anything.

## Counting to 2000

### BEFORE THE DAY — FOLLOW

#### **Lindqvist walks the drill floor round once**

The drill has a rhythm and it is not written down anywhere: which valve is checked at which point in the run, which noise means the barrel is loading. Lindqvist, the drill engineer, walks it once. Stay with him and stay out of the winch path, which is the one part of this camp that will not stop for you.

### PLAN CARD 96W

Wednesday, and the chronology is the thing holding up everything else. Okonkwo has counted to 1,180 metres by eye and the layers are getting thinner every hundred metres. Halvorsen wants the season's depth-age scale by the aircraft window, and the whole comparison with the Skarv record rests on it. Today you find where counting stops being possible, decide which annual signal to count with, and put a number on how deep the firm's own clock runs. Below the depth where layers stop resolving, every age comes from a model, and a model is not a count.

### BRIEFING 16W

Okonkwo needs the depth at which the layer thickness drops below what the line can resolve.

### WHAT YOU DECIDE 14W

Establish how a year is counted, and find the depth where counting stops working.

### 3.1 The depth the counting runs out

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · SWEEP

#### SCENE 55W

Snow buries snow, and each year's layer is squeezed thinner by the weight above it and the ice spreading below it. The scanner can separate two bands three millimetres apart and no closer. Okonkwo needs the depth where the annual layers thin to that limit, because past it the years cannot be counted at all.

#### GUIDE 59W

Drag the depth. The readout is the measured thickness of one year's layer there, in millimetres, and only the depths you stop at get plotted. You are not looking for a peak: you want the depth at which the thickness falls to the scanner's three-millimetre limit. Sweep down the core, find that crossing, then mark the depth and commit.

#### ASK 22W

The line resolves bands three millimetres apart and no closer. At what depth does the annual layer thickness fall to that limit?

#### BEHIND THE BUTTON 158W

Why layers thin with depth. Ice under ice is compressed, and an ice sheet also spreads sideways under its own weight, stretching each buried layer thinner as it goes. Near the surface a year is centimetres of ice; two kilometres down the same year's snowfall may be a few millimetres.

What the limit costs. Above the crossing, a year is a band somebody can point at, and the age of the ice is a count. Below it, two years arrive inside one scanner resolution and the count stops. The ages then come from a flow model instead, which is an inference with its own uncertainty rather than a tally.

Why the crossing is a depth rather than a year. The thinning depends on the ice's own flow, so the same limit lands at different depths in different cores. It has to be measured in each one, which is why this reading is taken before any chronology is published.

#### TAKES AS READ

layers are stretched thinner as the ice below them spreads outwards · accumulation rate, and what a site with little snow buys and costs — taken as read

#### VERDICT 95W

Ice spreads outwards under its own weight, so every layer is stretched thinner as it sinks, and the thinning compounds with depth. At Vestri the surface lays down about 12 centimetres a year in ice equivalent; 2 kilometres down the same year is a few millimetres. Where the layers close to within the resolution of the measurement, the count stops — not because the years are gone, but because two of them can no longer be told apart. Everything deeper is dated by a flow model anchored to whatever dated events can still be found.

#### TAKEAWAY 13W

A count ends where the instrument stops separating one year from the next.

### 3.2 Which year marker survives the depth

COLD LABORATORY · A ROOM · CHOICE

#### SCENE 27W

The isotope chemist, Yuki Tanabe, has 4 annual signals measured on the same stretch of core at 1,400 metres, where a year is about 26 millimetres thick.

#### GUIDE 61W

Four annual signals, and a year here is twenty-six millimetres thick. Ask of each whether the thing being measured can move through the firn. Water molecules travel through open pores for decades, and that smooths anything thinner than about twice the distance they cover. Sea salt and mineral grains do not travel. And two of these fade for other reasons entirely.

#### ASK 15W

At 1,400 metres, which signal is still able to separate 1 year from the next?

#### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

#### TAKES AS READ

water molecules can move through the connected air spaces in firn · firn: snow densifying into ice, and the close-off depth — taken as read · stable isotopes as a thermometer:  $\delta^{18}\text{O}$  and deuterium — taken as read

#### VERDICT 84W

Water molecules move through the open pores of the firn for decades. That movement smooths the isotope record. Anything thinner than about twice the diffusion length is averaged away, and at this site that removes the yearly signal long before 1,400 metres. Sodium arrives as sea salt. Dust arrives as mineral grains. Neither travels through the pore space, so their yearly structure survives as far down as the measurement can read it. Visible layering fades once the bubbles are gone, and conductivity marks eruptions.

#### TAKEAWAY 17W

A signal is only useful for counting while it is still sharper than the layer it marks.

### 3.3 How old the bottom of the firn is

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · BALLPARK

#### SCENE 31W

Brandt has the firn temperature string reading down to 88 metres, which is where the pores seal at this site. The surface is adding about 0.32 metres of snow a year.

#### GUIDE 63W

Four numbers, and two of them belong to other questions: the ice-equivalent rate, and the density of solid ice. Ask of each whether it describes snow as it lies. In the firn the layers have not been thinned by flow, so this is a straight division. Deeper down the same sum has to be done interval by interval, because the layers get thinner.

#### ASK 10W

Estimate the age of the snow at the close-off depth.

#### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

#### TAKES AS READ

a depth divided by an annual layer thickness gives a number of years · why ice keeps a record at all: burial without melting — taken as read · accumulation rate, and what a site with little snow buys and costs — taken as read

#### VERDICT 89W

A depth–age scale is arithmetic on layer thicknesses: divide the depth by the thickness one year adds and you have the years it took to get there. In the firn the layers have not yet been thinned by flow, so the sum is a straight division; deeper in the ice the same calculation has to be done interval by interval as the layers get thinner. Nearly three centuries of snow sit above the close-off depth at Vestri, and every one of those years is still open to the atmosphere.

#### TAKEAWAY 12W

A depth becomes an age only through the rate that filled it.

### 3.4 A year, or a summer that melted

CHRONOLOGY & THE CORE LINE · CORE LINE · A PERSON · BELT

#### SCENE 41W

Sections come out of the trench faster than they can be logged properly. A layer that formed over a year is one thing. A layer where the surface melted and refroze is another, and both are visible on the light table.

#### GUIDE 60W

Two bins, and the question is whether the layer is a year of accumulation or a melt event inside one. An annual layer pairs a coarse summer with a fine winter and keeps its bubbles. A melt layer is clear, dense and bubble-poor, because the water filled the spaces before it refroze. Sort on the ice, not on the label.

#### ASK 12W

Send each section to the bin that says what made the layer.

#### BEHIND THE BUTTON 93W

Why the difference decides the chronology. The count of annual layers is the clock. A melt layer counted as a year makes the core older than it is at every depth below; a year missed makes it younger, and both errors propagate all the way down.

What melt does to everything else. Percolating water moves the chemistry down through the layers it passes, so a melt layer is not only a counting problem — it smears the record either side of it, which is why the site was chosen where melt is rare.

#### TAKES AS READ

snow builds up in layers and can melt and refreeze

#### VERDICT 171W

An annual layer is built by a season: coarse crystals from summer sun, fine wind-packed snow through winter, and air trapped as bubbles throughout. A melt layer is built by water. The surface thaws, the water percolates down and refreezes in the pore space, so what comes out on the light table is clear, dense and almost free of bubbles. Both are stripes in a core and only one of them is a year. The cost of confusing them is not local: annual layer counting is the clock, so a melt layer counted as a year makes every date below it too old, and a year missed makes every date too young, all the way to the bottom of the core. Melt also does something worse than miscount. Water moving down carries the chemistry with it, so the isotopes and the ions either side of the layer are smeared — which is why a site is chosen for how rarely its surface melts rather than for how much snow it gets.

#### TAKEAWAY 15W

A layer counted wrong moves every date below it, in the same direction, for ever.

### 3.5 Why the hole is on the dome

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · CALLBACK · CHOICE

#### SCENE 37W

The drill engineer, Pär Lindqvist, has the hole at 2,040 metres and a map on the wall with 3 abandoned sites on it, all of them on slopes. Vestri Dome is the high point for 200 kilometres.

#### GUIDE 69W

Four things a dome might buy, and three of them are about the ice being cold, thick or flat. Ask of each whether it is about where the ice came from. An ice sheet flows outward from its high points. On a slope the ice at a given depth arrived from upstream, so the record mixes one place's weather with a journey. Domes pay for that in thin layers.

#### ASK 14W

What does drilling at the dome buy that a site on the slope cannot?

#### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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#### TAKES AS READ

ice spreads outwards from the high points of an ice sheet · why ice keeps a record at all: burial without melting — taken as read

#### VERDICT 88W

An ice sheet flows outwards from its high points. On a slope the ice at a given depth arrived from somewhere upstream, so a core there mixes the weather of one place with the history of a journey, and the two cannot be separated afterwards. Under a dome the flow is almost straight down, so the column below the drill is very nearly the snow that fell on it. Domes pay for that: they are high, cold and dry, which means thin layers and a long slow season.

#### TAKEAWAY 18W

A core is a column of what fell above it only where the ice has not travelled sideways.

#### END OF THE DAY 18W

Every dating method has a depth where it stops working, and knowing it is part of the date.

## One aircraft, and what leaves on it

### PLAN CARD 99W

Thursday is the aircraft window and 11 Skarv samples are in the core store until it opens. Halvorsen has the Skarv record on one screen and this station's on the other, and they disagree by more than either group's stated uncertainty. She wants a measurement made while both cores can be run on one instrument. Okonkwo wants nothing said until the timescales are matched. Today you choose what to measure before the plane leaves, what running both on the same machine settles, and what it does not. After Thursday the comparison is between two laboratories rather than two cores.

### BRIEFING 12W

The Skarv group's samples fly out on Thursday and cannot come back.

### WHAT YOU DECIDE 13W

Decide what is measured while the Skarv samples are still on the station.

## 4.1 What is worth measuring before Thursday

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · VALUE

### SCENE 36W

11 Skarv samples, 1 working day, and the isotope line can take 4 of them. The data manager, Jonas Aalto, has the two records side by side and the difference between them on a third screen.

### GUIDE 49W

Eleven samples, one working day, and the isotope line takes four. Open each candidate and ask what its result would change about the difference between the two records. Four runs that confirm what is already agreed have spent the day; four that could move the disagreement have bought something.

### ASK 7W

Which four runs should the day buy?

### BEHIND THE BUTTON 137W

Why the difference is the subject. Two records side by side, with a gap between them on a third screen, is the whole question of the week. A run that tightens a part of the record nobody disputes cannot narrow that gap.

What a day of the isotope line is worth. It is the only instrument that can settle certain questions and it runs four samples. That makes each slot a choice against the other seven, and the deadline is Thursday rather than the end of the season.

Why sampling depth matters as much as sample count. Four runs clustered in one interval answer one question well. Four spread across the disagreement answer a different question — which of them is worth more depends on whether the gap is thought to be an offset or a shape.

### TAKES AS READ

two laboratories measuring the same quantity can differ by a fixed amount

### VERDICT 91W

Two laboratories measuring the same water can differ by a fixed amount, and a fixed amount looks exactly like a climate difference when it is plotted against depth. The standards measure that offset directly; the overlapping depths test it on real ice. Between them they either remove the explanation or establish it. The deepest sample is interesting and settles nothing, since nothing here overlaps it, and the density profile is already in the other group's published report. Repeating our own sample tests our line, which is not the thing in dispute.

### TAKEAWAY 16W

A measurement is worth its cost when the decision changes depending on how it comes out.

## 4.2 The same water, twice

COLD LABORATORY · A PERSON · CHOICE

### SCENE 31W

Tanabe has both groups' working standards on the bench and 30 minutes of instrument time. Skarv reports its values against its own standard, calibrated in a different laboratory four years ago.

### GUIDE 71W

Four things a joint run might establish. Ask of each whether one instrument in one afternoon could separate it from the alternatives. An isotope value is a difference from a reference water, so it carries whatever that reference is worth. Two laboratories keep their own. A scale gap and a real site difference both appear as a constant shift. That is why one of these needs both standards on one instrument.

### ASK 9W

What does running both standards on this instrument establish?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

an isotope value is reported as a difference from an agreed reference water

### VERDICT 91W

An isotope value is not an amount; it is a difference from a reference water, so it carries whatever that reference is worth. Two laboratories keep their own working standards, both traceable to the same international water and both slightly off it. Run side by side on one instrument, the two standards show the gap between the scales directly, in the same units the records are plotted in. Nothing else on the bench can separate that gap from a real difference between the sites, because both appear as a constant shift.

### TAKEAWAY 15W

A ratio reported against two different references is two different numbers about the same water.

### 4.3 Which differences a shared instrument removes

TRAPPED GAS · GAS LABORATORY · A ROOM · PROTOCOL

#### SCENE 30W

The gas geochemist, Samuel Adeyemi, has four ways the two records could differ written on the board, and one working day in which both cores are in the same building.

#### GUIDE 62W

Four ways the records could differ, and four kinds of evidence. Pair them by asking what sharing an instrument can and cannot remove. A stretched timescale is a chronology problem and needs an event dated in both cores. A real climate difference is what survives after everything else is eliminated. And melting alters the outside of a section more than the inside.

#### ASK 9W

Sort what a joint run can and cannot settle

#### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

#### TAKES AS READ

two measurements of one quantity can differ for reasons that have nothing to do with the quantity · accumulation rate, and what a site with little snow buys and costs — taken as read · annual layer counting, and where counting fails — taken as read

#### VERDICT 90W

Only the first of these is fixed by sharing an instrument. A stretched timescale is a chronology problem and needs an event of known date present in both cores. A real climate difference is what is left once every other explanation has been removed, which is why it is established by elimination rather than measured directly. Melting in transit alters the outer ice more than the inner, so it leaves a signature in the

profile across a section rather than in the average. Four differences, four different pieces of evidence.

**TAKEAWAY 13W**

Sharing an instrument removes the differences the instrument was causing, and only those.

## 4.4 How thick a year is here

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · CALLBACK · BALLPARK

### SCENE 32W

Brandt has the pit wall photographed and the last summer surface marked at 32 centimetres down. The snow above it weighs a third of what the same thickness of solid ice would.

### GUIDE 68W

Five numbers, and two of them belong elsewhere: the density of liquid water, and the days of stake readings. Ask of each whether the conversion needs it. Snow and ice are the same material at different densities, so a thickness of snow is not a thickness of record. Only ice equivalent lets two sites be compared, and this site gets a tenth of what a coastal one does.

### ASK 7W

Estimate this year's accumulation in ice equivalent.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

snow becomes ice by being compressed under the snow that falls on it ·  
accumulation rate, and what a site with little snow buys and costs — taken as read

### VERDICT 102W

Snow and ice are the same material at different densities, so a thickness of snow is not a thickness of record. Multiplying the layer by the ratio of the two densities converts it to the ice it will become once the air is squeezed out, which is the only form in which two sites can be compared. Vestri gets about 12 centimetres a year in ice equivalent. A coastal site gets a metre, and that difference decides everything downstream: how far back the core reaches, how thin the layers get, and how much older the ice is than its own trapped air.

### TAKEAWAY 15W

An accumulation rate means nothing until it is stated as the ice it will become.

### END OF THE DAY 16W

Some comparisons are only available for as long as both things are in the same room.

## Turning a ratio into a temperature

### PLAN CARD 93W

Friday, and the isotope profile is complete to 1,600 metres. Tanabe has the numbers and Halvorsen wants them in degrees for the season summary, which means a slope, and the slope is a measurement somebody made rather than a constant. Brandt has 11 years of mast temperatures to test it against. Today you convert the profile into a temperature change, decide how tightly the conversion can be stated, and settle what the isotopes are actually a thermometer of. A slope taken on faith turns every degree in the record into somebody else's assumption.

### BRIEFING 17W

The isotope profile is finished to 1,600 metres and nobody has agreed what it says in degrees.

### WHAT YOU DECIDE 15W

Establish what the isotope record measures and how confidently it can be stated in degrees.

## 5.1 What the ratio says in degrees

COLD LABORATORY · A ROOM · BALLPARK

### SCENE 32W

Tanabe has the isotope profile against depth, and the section covering the last cold interval sits 1.8 per mil below the modern surface value. The site's calibration slope is on the board.

### GUIDE 58W

Four numbers, and two of them belong elsewhere: a slope for a different isotope, and the years of mast overlap. Ask of each whether this conversion uses it. And note where the slope comes from. It is not a constant of physics but a fit to this site's own weather, measured over the years that have both records.

### ASK 8W

Estimate the temperature change that isotope shift represents.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

colder air loses its heavier water molecules earlier, leaving the snow isotopically lighter · stable isotopes as a thermometer:  $\delta^{18}\text{O}$  and deuterium — taken as read

### VERDICT 87W

Water carrying the heavy isotope condenses first. Air that has cooled further has already lost more of it, so the snow falling last is the lightest. Over a useful range the ratio and the temperature move together in a straight line. The slope of that line is not a constant of physics. It belongs to this site's weather, and it is measured by comparing the isotope record with the mast during the years that have both. Dividing the shift by the slope gives the change in degrees.

### TAKEAWAY 15W

A slope converts a measurement into a claim, and it is measured rather than assumed.

## 5.2 How firmly the cooling can be stated

SURFACE & ACCUMULATION · SNOW STUDY HUT · A PERSON · CLOUD

### SCENE 36W

Brandt has the mast record, the calibration years and the scatter around the fitted slope. The season summary needs a statement that the interval was colder than the present, and how sure of it anybody is.

### GUIDE 63W

The band is the reconstructed cooling with its calibration uncertainty on it. The summary wants to claim the interval was more than two degrees colder. Two pieces of work are available and they act differently on the band: one moves it, one narrows it. Apply what you would fund, watching the middle and the width separately. Then say whether the claim is cleared.

### ASK 27W

Which work would let the summary state that the interval was more than two degrees colder? Place the bars to report the mean and its one-sigma uncertainty.

### TAKES AS READ

a fitted relationship carries a spread as well as a central value · random against systematic uncertainty — taken as read · separating a trend from year-to-year noise — taken as read

### VERDICT 88W

The reconstruction is a distribution, not a number: the slope was fitted to 11 years of overlap and it carries the scatter of that fit. A claim that the interval was more than two degrees colder is a claim about where the whole distribution sits, so it is cleared by making the distribution narrower rather than by moving its middle. More overlap years is the only action here that adds information; the source-region correction changes the central value and leaves the spread exactly as wide as it was.

### TAKEAWAY 14W

Narrowing a distribution and moving it are different actions with different evidence behind them.

### 5.3 The temperature of what, and where

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · CHOICE

#### SCENE 29W

Aalto is drafting the summary line and has written that the isotopes give the temperature of the last cold interval. Halvorsen wants the sentence narrowed before it goes anywhere.

#### GUIDE 66W

Four things the isotope record might be about. Ask of each where and when the ratio is actually set. It is fixed as vapour condenses, in the cloud, in the seasons that deliver most of the snow. Getting from there to a ground temperature needs the site calibration. Two of these options are separate measurements entirely, one of them taken with a thermometer in the hole.

#### ASK 8W

What does the isotope record most directly record?

#### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

#### TAKES AS READ

snow forms in a cloud and falls to a surface some distance below it

#### VERDICT 91W

The isotope ratio is set as water vapour condenses, so it records conditions in the cloud that produced this snow, weighted towards the seasons that deliver most of it. Getting from there to a surface temperature needs the relationship between cloud and ground, which is what the site calibration supplies. The ocean the water came from does leave a mark, which is why source-region corrections exist. The ice's own temperature is a separate measurement, taken with a thermometer in the borehole, and it says nothing about the year the snow fell.

#### TAKEAWAY 13W

A proxy measures one specific thing, and the sentence has to say which.

## 5.4 What makes one year different from the next

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CALLBACK · PROTOCOL

### SCENE 33W

A metre of core from 60 metres is on the light table. The chronology lead, Ines Okonkwo, has four features marked on it and wants each one attributed before the section is cut.

### GUIDE 63W

Four features on one metre of core, and four things that could have made them. Pair them by asking what each feature is physically made of: crystals, acid, mineral grains, or bubbles. Then ask what season each belongs to. No single marker survives the whole core, so a year is where several independent signals agree. Where only one does, it is a candidate.

### ASK 6W

Read a year off the core

### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

### TAKES AS READ

snow that falls in summer sits under different conditions from snow that falls in winter

### VERDICT 86W

No single marker survives the whole core, so a year is established where several independent signals line up. Coarse crystals form on a summer surface that stayed exposed; conductivity spikes are acid, and volcanic acid arrives in a season; dust is delivered by winds thousands of kilometres away and peaks with their season; the light and dark banding is bubble structure changing as each layer was compressed. Where three of the four agree, a year is a year. Where only one does, it is a candidate.

#### TAKEAWAY 15W

A year is counted from several signals that agree, not from one that looks convincing.

#### END OF THE DAY 16W

A proxy becomes a number through a calibration, and the calibration is part of the result.

## Air that stopped moving

### BEFORE THE DAY — HUNT

#### Six core boxes on the manifest, four in the trench

The aircraft takes what is boxed and labelled, and two boxes from last week are not where the manifest says. Until they are found nobody can say what is flying out and what is staying another year, and ice that stays another year is ice with another year's temperature history on it.

### PLAN CARD 103W

Saturday, and Adeyemi has carbon dioxide from four depths and a problem with all of them. The air in a bubble stopped moving centuries after the snow around it fell, and at a site this dry the difference is the largest correction in the whole record. Okonkwo needs it before the two timescales can be put together. Today you work out how large the offset is here, name the step in the chain that decides it, and settle why the gas record is smoother than the ice one. A gas record plotted on the ice's own clock is a record shifted by centuries.

### BRIEFING 16W

The gas record and the ice record from the same depth are not the same age.

### WHAT YOU DECIDE 17W

Establish what a bubble is, when it sealed, and why its air is younger than its ice.

## 6.1 How much younger the air is

TRAPPED GAS · GAS LABORATORY · A ROOM · BALLPARK

### SCENE 36W

The pores seal at 88 metres here. Above that the firn is open to the atmosphere, and its mean density is about 0.62 of solid ice. Adeyemi has the accumulation rate on the board beside it.

### GUIDE 69W

Four numbers, and one of them is the same year written as a snow thickness rather than as ice. Ask of each whether the units match the others. The air keeps mixing with the atmosphere until the pores close. So the bubble is as old as the sealing, and the ice is as old as the snow. Get this wrong and a gas record is plotted centuries too early.

### ASK 12W

Estimate the age difference between a bubble and the ice around it.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

air moves freely through connected pore spaces until they close

### VERDICT 101W

Air circulates through the connected pore spaces of the firn and mixes with the atmosphere until those pores close, so the air in a bubble is roughly as old as the moment of sealing while the ice around it is as old as the snow. Converting the firn column to ice equivalent and dividing by the accumulation rate gives that interval. It is set by how deep the close-off is and how fast snow arrives, so cold dry sites carry offsets of centuries and wet coastal ones a few decades. Ignore it and a gas record is plotted centuries too early.

### TAKEAWAY 18W

A bubble is sealed at a depth, and the ice around it arrived at that depth long before.

## 6.2 From snowfall to sealed bubble

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · CHAIN

### SCENE 26W

Lindqvist has the borehole's own record on the wall: five stages between a snowflake landing and air being locked away, with a measurement against each one.

### GUIDE 55W

Put the five stages in the order a snowflake actually passes through them, from landing to being sealed in. Then name the stage that decides how much older the ice is than the air inside it. Each stage has a measurement against it, so read those rather than choosing the stage that sounds most important.

### ASK 14W

Which stage decides how much older the ice is than the air inside it?

### BEHIND THE BUTTON 166W

Why the ice and its air have different ages. Fresh snow is open to the atmosphere, and it stays open as it compacts. Air moves in and out of that layer for decades or centuries. Only when the pores finally pinch off is the air sealed, and by then the ice around it has been buried a long time.

What sets the size of the gap. The depth at which the pores close, and how long the snow takes to get there. Both depend on temperature and on how much snow falls each year, which is why the gap is hundreds of years at a cold, dry site and much less at a warm, snowy one.

Why it matters more than it sounds. Every comparison between a gas record and an ice record has this offset in it. Read a carbon dioxide rise against a temperature change without correcting for it and you can get the order of events wrong, which is a claim about cause.

### TAKES AS READ

a quantity that passes through several stages is limited by one of them

### VERDICT 93W

Every stage moves something, but only one of them fixes the size of the offset. Air exchanges with the atmosphere all the way down the firn, so its age barely changes while the ice around it gets older year by year; the moment the pores seal is the moment that exchange stops, and the age difference is fixed at whatever the ice's age is at that depth. Mixing looks like the answer because it is the process that keeps the air young, but its effect is a decade against a difference of centuries.

### TAKEAWAY 16W

In a chain of stages, one of them sets the answer and the others follow it.

### 6.3 Why the gas record has no sharp edges

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CHOICE

#### SCENE 35W

Okonkwo has the two records from the same depths on one plot. The ice record has features a few years wide. The gas record from the same interval has nothing narrower than a few decades.

#### GUIDE 66W

Four explanations for a record with no sharp edges. Ask of each whether it is a property of the site or of the laboratory. Cutting smaller pieces would help one of them a little. The pores do not all close at once, so the air sealed in one layer is a mixture spanning the years that range covers. A wetter site closes them in a decade.

#### ASK 14W

Why is the gas record smoother than the ice record from the same depths?

#### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

#### TAKES AS READ

a measurement that averages over an interval cannot show anything shorter than it · stable isotopes as a thermometer:  $\delta^{18}\text{O}$  and deuterium — taken as read

#### VERDICT 87W

Air is in contact with the atmosphere throughout the open firn, and the pores do not all close at once — the closing happens across a depth range, so the bubbles sealed in one layer hold a mixture spanning the years that range represents. That smoothing is a property of the site: a high-accumulation site closes its pores in a decade and keeps sharp features, while this one takes several. Sample size makes a small contribution and could be reduced by cutting smaller pieces. The smoothing cannot.

#### TAKEAWAY 15W

A record made by an averaging process cannot show a change faster than the averaging.

## 6.4 The sample that was clean until it was handled

COLD LABORATORY · A ROOM · CALLBACK · DIAGNOSIS

### SCENE 33W

The cold laboratory technician, Elena Cruz, has a sodium result four times higher than the run before it. She has the blank, the run order and the section's handling record on the bench.

### GUIDE 65W

Five readings, and the quiet ones do the work. The blank before is ordinary and the blank after is high. The trim was done to specification. The replicate core holds the same year with no sodium in it. Ask of each candidate how many of the five it covers. What identifies the source is not the loud number but which of the others stayed quiet.

### ASK 7W

Find out where a signal came from

### BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

### TAKES AS READ

a blank is run through the same process as a sample, with no sample in it · what a proxy is: the chain between the thing and the number — taken as read

### VERDICT 86W

The blank before the sample is ordinary and the blank after it is high, which places the contamination inside the run rather than in the ice. The trim was done to specification, so the outside of the section is ruled out, and the replicate core holds the same year with no sodium in it, so the atmosphere is ruled out as well. Every clean reading here is doing work: what identifies the source is not the loud number but which of the quiet ones stayed quiet.

### TAKEAWAY 15W

A blank measures the process, so anything it shows was never evidence about the ice.

#### END OF THE DAY 19W

Two things in the same piece of ice can be different ages, and the difference has to be calculated.

## What the wind brought

### PLAN CARD 100W

Sunday, and the chemistry line has run the 1,100 to 1,300 metre section. Dust is up by a factor of four through the cold interval, sodium is up by half, and there is one sulphate spike that Okonkwo wants identified before it is used for anything. Cruz has the isotope ratios from the same section. Today you turn a raw ratio into an isotope value, say what the dust is evidence of, and decide which chemistry the season should pay to measure. Chemistry that arrives from somewhere else is the only part of this record that is about anywhere else.

### BRIEFING 15W

Dust, sea salt and volcanic acid arrive from three different places for three different reasons.

### WHAT YOU DECIDE 15W

Use the chemistry in the ice to date it and to say something about elsewhere.

## 7.1 The number that is a difference

COLD LABORATORY · A ROOM · BALLPARK

### SCENE 35W

Cruz has the raw ratio of heavy to light oxygen for a sample and the same ratio for the standard water. The record is plotted in parts per thousand and the raw numbers are not.

### GUIDE 62W

Four numbers, and one of them is the site temperature slope, which belongs to a later step. Ask of each what part of the convention it supplies. Ratios differ in the fourth decimal place, so nobody reports them as ratios. The scale is a comparison with one agreed water, set so that the standard itself reads zero. Polar snow is always negative.

### ASK 11W

Estimate the isotope value of this sample in parts per thousand.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

an isotope ratio is the amount of a heavy isotope divided by the amount of a light one

### VERDICT 98W

Isotope ratios differ from one another in the fourth decimal place, so nobody reports them as ratios. The convention is to divide the sample's ratio by the ratio of 1 agreed water, subtract 1 so that the standard itself reads 0, and multiply by 1000 so the numbers are convenient. What comes out is a difference in parts per thousand, always negative for polar snow because every stage of the journey from the ocean has already removed some of the heavy isotope. The scale is a comparison, and it only means anything if everyone uses the same water.

### TAKEAWAY 20W

An isotope value is a comparison with an agreed water, which is why every one of them is negative here.

## 7.2 Four times the dust

CHRONOLOGY & THE CORE LINE · CORE LINE · A PERSON · CHOICE

### SCENE 37W

Through the cold interval the dust concentration is four times the modern value and the grains are the same size distribution as today's. Okonkwo has the sodium record beside it, up by half over the same depths.

### GUIDE 63W

Four readings of four times the dust, and three of them name a cause. Ask what a concentration is made of. It is an amount of dust divided by an amount of snow. Three separate things move it: how much was released, how well it was delivered, and how much snow fell to dilute it. Accumulation drops in cold intervals on its own.

### ASK 11W

What is the most defensible reading of four times the dust?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

fine dust reaching this plateau has travelled thousands of kilometres

### VERDICT 89W

A concentration is an amount of dust divided by an amount of snow, so it responds to three separate things: how much dust the source regions released, how efficiently the winds delivered it, and how much snow fell here to dilute it. Accumulation at this site drops in cold intervals, which raises concentration on its own. Separating the three needs the flux — concentration multiplied by accumulation — and the grain size and chemistry to identify the source. The concentration alone establishes that the delivery changed, and no more.

### TAKEAWAY 21W

A concentration in the ice is a product of what was produced, what was carried, and how much snow diluted it.

### 7.3 What the chemistry budget should buy

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · SCIENCETANK

#### SCENE 24W

Halvorsen has one chemistry line, six weeks, and four proposals from three groups. The depth–age scale below 1,600 metres is the season's stated objective.

#### GUIDE 85W

The section below 1,600 metres carries no dated horizon and the layer count stopped above it. Two eruptions of known date are expected somewhere in that depth range. Sulphate and conductivity find candidate horizons; tephra shard chemistry is what puts a name and a date on one. The modern pollution record is a genuine result and bears on nothing in the timescale. The sodium section has been measured twice and both runs agree inside the analytical error. There is one chemistry line and six weeks.

#### ASK 7W

Fund the chemistry that dates the core

#### BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

#### TAKES AS READ

an eruption of known date leaves a signature in ice all over the world · random against systematic uncertainty — taken as read · separating a trend from year-to-year noise — taken as read

#### VERDICT 74W

The season's stated objective is a depth–age scale below the counted section. Only the volcanic work bears on it. Sulphate and conductivity find the candidates. The tephra chemistry turns a candidate into a named eruption with a date on it. Neither is worth much without the other. The pollution record is a real result for a different question. And a third run on ice that already has two agreeing measurements buys nothing at all.

#### TAKEAWAY 22W

Work that could change what the season can claim is worth more than work that adds detail to what it already claims.

## 7.4 The run through the brittle zone

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · CALLBACK · SEQUENCE

### SCENE 39W

The barrel is at 780 metres, inside the depth range where the ice comes up cracked. The winch, the fluid and the storage are all set for ordinary ice and Lindqvist wants the order settled before the lift starts.

### GUIDE 64W

All five of these will happen, so ask what each one buys the next. Between about five hundred and twelve hundred metres the bubbles are at pressures the surface cannot support. Ask of each step whether it reduces the shock, lets the pressure equalise, or fixes information that is about to be lost. One of them destroys the depth record if it comes early.

### ASK 6W

Bring a brittle section up whole

### BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

### TAKES AS READ

gas trapped at depth is at the pressure of the ice around it · why ice keeps a record at all: burial without melting — taken as read

### VERDICT 83W

Between about 51200 metres the bubbles are at pressures the surface cannot support, and the ice around them fractures as it is brought up. Each step buys the next one: a slow lift reduces the shock, resting the core at hole temperature lets the pressure equalise gradually, logging the breaks fixes what depth each fragment came from before anything moves, and storing it for weeks lets it finish cracking. Cutting first destroys the depth information and cuts a section that is still moving.

### TAKEAWAY 11W

Ice recovered badly is a shorter record, not a cheaper one.

What the wind carried is evidence about the places it came from, not only about here.

## The year that is written everywhere

### PLAN CARD 113W

Monday, and the chemistry has found a sulphate spike at 1,642 metres with glass shards in it. If it is the eruption Okonkwo thinks it is, the deep section gets its first fixed date and the two records get their first common point. Adeyemi wants to know what it does for the gas record, and the weather forecast gives Brandt one clear day for three tasks. Today you identify the spike, say what a tie point fixes, and choose what gets done before the wind comes back. A horizon named wrongly is worse than none: every age below it inherits the mistake, and the record ends up looking better dated than it is.

### BRIEFING 14W

A sulphate spike at 1,642 metres could be the anchor the deep section needs.

### WHAT YOU DECIDE 19W

Use a dated eruption to tie this core to another one, and know what that tie does not fix.

## 8.1 The spike at 1,642 metres

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · DIAGNOSIS

### SCENE 35W

One sulphate spike, two candidate eruptions of similar age, and a set of glass shards recovered from the same centimetre. Okonkwo has the shard chemistry, the layer count and the modelled age on the bench.

### GUIDE 59W

Five readings, and only one of them is independent of the timescale. Ask of each whether it could tell two eruptions ninety years apart from each other. The modelled age cannot: its own uncertainty is wider than the gap. Sea-salt sulphate does not arrive with volcanic glass. And the layer count stopped four hundred and sixty metres higher up.

### ASK 3W

Identify the horizon

### BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

### TAKES AS READ

volcanic glass carries the chemical composition of the volcano that erupted it

### VERDICT 88W

The shard chemistry is the only reading here that is independent of the timescale, and it is the one that discriminates: glass composition is a fingerprint of a particular magma and a particular volcano. The modelled age cannot choose between 2 candidates 90 years apart because its own uncertainty is larger than that gap. Sea-salt sulphate does not arrive with volcanic glass. And waiting for the layer count is waiting for something that stopped 460 metres higher, which is the reason a dated horizon matters here at all.

### TAKEAWAY 21W

A horizon is identified by what is in it, not by the date it would be convenient for it to have.

## 8.2 What a dated ash layer fixes

TRAPPED GAS · GAS LABORATORY · A PERSON · CHOICE

### SCENE 24W

Adeyemi has the eruption date, the depth it sits at, and the carbon dioxide measurements from the bubbles in the same section of core.

### GUIDE 69W

Four things a dated ash layer might fix. Ask of each which of the two clocks in the core it belongs to. Ash falls on the surface and is buried with the snow. The air beside it was sealed centuries later and lower down. So a tie point fixes one clock and leaves the other to a calculation. That is why the age difference has to be estimated first.

### ASK 6W

What does the dated horizon fix?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

volcanic ash settles out of the atmosphere within a year or two of an eruption

### VERDICT 89W

Ash falls onto the surface and is buried with the snow, so it dates the ice it sits in. The air in the bubbles beside it was sealed centuries later, at the close-off depth, and it is that much younger. A tie point therefore fixes one of the two clocks in the core and leaves the other one to be reached by calculation — which is exactly why the age difference has to be estimated before any comparison between a gas record and an ice record can be made.

### TAKEAWAY 19W

A tie point dates the material it is in, and anything else has to be reached by a calculation.

### 8.3 One clear day, three jobs

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · TRIAGE

#### SCENE 30W

Brandt has a forecast that gives her one clear day and then four of blowing snow. Three jobs are outstanding and each takes most of a day on its own.

#### GUIDE 59W

Four jobs, all of them a day's work. Ask of each whether it could be done in four days of blowing snow, and whether the storm changes what it measures. The pit is beside the hut and the mast is on a fixed line. One of these runs two kilometres out and measures a surface that drifting snow rearranges.

#### ASK 6W

Which job takes the clear day?

#### BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

#### TAKES AS READ

a measurement that cannot be repeated should be taken when it can be taken

#### VERDICT 90W

Two of these can be done in bad weather and one cannot. The pit is beside the hut and the mast is a short walk on a fixed line, so both survive four days of blowing snow with nothing worse than discomfort. The stake array runs two kilometres out and needs visibility to level against; more to the point, blowing snow redistributes the surface the stakes are measuring, so a reading taken after the storm is measuring something different. The shallow core can be re-drilled in any weather at all.

#### TAKEAWAY 13W

Order the work by what becomes impossible, not by what is most important.

## 8.4 The level that holds the hole open

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · CALLBACK · HOLD

### SCENE 43W

The borehole is three kilometres deep and the ice around it is trying to close it. The fluid column is what holds it open, and the level in the tower has to stay where the pressure balance needs it while the drill runs.

### GUIDE 58W

Hold the fluid level inside the band on the tower gauge. The band narrows as the hole deepens, because the deeper the hole the less margin there is between closing it and fracturing it. The pump between the tank and the hole is your control, and every run of the winch changes the balance until you answer it.

### ASK 8W

Hold the fluid level while the drill runs.

### BEHIND THE BUTTON 142W

Why the hole needs holding at all. Ice at three kilometres is under enough pressure to flow, and it closes a borehole slowly and permanently. A column of fluid at close to the ice pressure holds it open, which is the only reason a deep hole can be re-entered at all.

Why too much is as bad as too little. Overpressure fractures the wall and takes fluid into the ice, which is both a lost hole and a contaminated record. The band is two-sided for that reason, and it is not symmetric in consequence.

Why the drill itself moves it. Pulling the drill out displaces less than it did going in, chips come up with fluid on them, and cold fluid coming off the reel is denser than what is in the hole. Each of them is a rate, not a step.

### TAKES AS READ

a column of fluid exerts pressure that grows with its depth · why ice keeps a record at all: burial without melting — taken as read

### VERDICT 174W

Ice this deep flows, and a borehole through it closes on its own unless something holds it open. That something is the fluid column, whose pressure at the bottom has to sit close to the pressure of the ice around it — a little under and the wall creeps in, a little over and it fractures and swallows fluid. Everything in a drilling run disturbs that balance and none of it is a single step. Pulling the drill removes displacement continuously while the winch runs; chips come up carrying fluid; cold fluid off the reel is denser than the column it joins and keeps sinking into it. So the pump is set to a new rate and held rather than tapped. The band narrows with depth because the two failure modes converge: the deeper the hole, the smaller the gap between the pressure that closes it and the pressure that breaks it, and both ends of that gap cost the same thing — a hole nobody can re-enter and a season nobody can repeat.

### TAKEAWAY 15W

A hole is held open by a balance, and a balance is held by somebody.

#### END OF THE DAY 17W

A dated event in two records fixes them to each other at that moment, and nowhere else.

## Where the record runs out

### BEFORE THE DAY — CANVASS

#### Who actually heard the generator stop

The number two generator stopped some time in the night and the log says nothing about it. How long the core store was without power decides what the season can claim. Ask round the camp until enough accounts agree on when the lights flickered.

### PLAN CARD 110W

Tuesday, and the drill has passed 2,400 metres. Lindqvist has clean core and Okonkwo has ages that no longer come from counting anything. Halvorsen wants the deep section in the season summary, which means saying how well it is dated, and Aalto has three uncertainties to combine before anybody writes a number down. Today you work out how thin a year is at that depth, decide which uncertainty is worth spending the season's remaining time on, and settle what the ice just above the bed can be used for. Deep ice is the whole reason the hole is here, and it is also the part with the least evidence attached.

### BRIEFING 16W

The drill is at 2,410 metres and every year down there is a few millimetres thick.

### WHAT YOU DECIDE 16W

Establish how thin the deep layers are and which uncertainty dominates what is claimed from them.

## 9.1 A year at 2400 metres

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · BALLPARK

### SCENE 34W

The ice is 3,010 metres thick here and the barrel is at 2,400. At the surface a year is about 0.119 metres of ice equivalent, and Lindqvist wants to know what 1 is now.

### GUIDE 65W

Four numbers, and one of them is the close-off depth, which belongs to the firn rather than to the flow. Ask of each whether it describes the surface, this depth, or the whole sheet. The thinning depends on how much ice is left below, so what matters is the distance to the bed. That is what packs a hundred thousand years into the bottom kilometre.

### ASK 10W

Estimate the thickness of 1 annual layer at 2,400 metres.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

ice spreads outwards under its own weight, stretching the layers below · annual layer counting, and where counting fails — taken as read

### VERDICT 88W

Ice flows outwards from the dome, and to keep going it has to stretch, so each layer thins as the weight above it grows. The simplest model treats the vertical strain as uniform, which makes a layer's thickness fall in proportion to the distance left to the bed: at four-fifths of the way down, a layer is one-fifth of what it was. That is what packs 100000 years into the bottom kilometre, and it is also why the count stops and the dating starts depending on a model.

### TAKEAWAY 14W

Thinning is why a deep core holds so much time in so little ice.

## 9.2 Which width the answer is made of

RECORDS & REPORTING · SCIENCE MODULE · A PERSON · PROPAGATE

### SCENE 31W

Aalto has the dust flux from the cold interval and four quantities behind it, each with its own uncertainty. There is time to improve one of them before the season ends.

### GUIDE 57W

The panel shows an error budget: each input's own width, times the power it carries in the calculation, contributes a bar. Read the bars rather than the inputs. There is time to improve one quantity before the season ends, so buy the measurement that shortens the tallest bar — not the one that is easiest to improve.

### ASK 7W

Which measurement should the remaining time buy?

### BEHIND THE BUTTON 129W

How the widths combine. In a product of quantities, the fractional uncertainties add in quadrature, each weighted by its exponent. A quantity raised to a power contributes that much more, and one measured badly can dominate even with an exponent of one.

Why the largest quantity is not the widest one. A big number measured precisely contributes little; a small one measured roughly can carry the whole budget. The bars are what separate those two, and the intuition to be resisted is the one that goes by the size of the number.

Why improving the wrong term is worse than doing nothing. It spends the last of the season and moves the total uncertainty by almost nothing, and it produces a report that looks more careful than it is.

### TAKES AS READ

a quantity multiplied into a result carries its own uncertainty into that result

### VERDICT 84W

A term enters the result raised to some power, and its contribution to the width of the answer is that power multiplied by its own fractional uncertainty. The grain diameter is cubed, which makes it look like the term to fix, but it is known to 5%, so it contributes 15. The accumulation rate enters once and is known to 30, so it contributes twice as much on its own. Ranking by exponent is a reasonable habit and it picks the wrong measurement here.

### TAKEAWAY 25W

A term's contribution is its own width multiplied by the power it is raised to, and the largest power is not always the largest contribution.

### 9.3 The last hundred metres

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CHOICE

#### SCENE 32W

Holt has core from the bottom of the drilled section. The layers are folded back on themselves in places and the isotope profile has reversals in it that no year could produce.

#### GUIDE 61W

Four options for folded basal ice. Ask of each what has been lost and what has not. The material is unchanged; the order is gone. So anything where depth means time is ruled out, and anything an unordered sample can answer is not. Note also what has to travel with the data, because the next person will not see the fold.

#### ASK 9W

What is the folded basal section still good for?

#### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

#### TAKES AS READ

ice near the bed has flowed over rough ground for a long time

#### VERDICT 99W

Ice near the bed has been dragged over rough ground for a long time, so layers are folded, sheared and sometimes overturned. The material is unchanged and the order is gone, which rules out anything that depends on depth meaning time — a temperature history, a layer count, a rate of change. It does not rule out what an unordered sample can still answer: what gases are present, how much air the ice holds, how old the oldest ice down there is. The disturbance has to travel with the data, because the next person will not see the fold.

#### TAKEAWAY 12W

Ice can be perfectly good material and still hold no readable order.

## 9.4 What the saw is cutting for

COLD LABORATORY · A ROOM · SPOT

### SCENE 44W

Sections come to the saw with a cutting list, and the list is set by whichever analysis has priority. The chief scientist changes that priority as the season goes — after the gas request, after the dating problem, after the aircraft date is confirmed.

### GUIDE 51W

Cut the sections the current priority wants and leave the rest in the rack. The priority on the cold-room board is changed during the day and nobody announces it. What is scored is the sections either side of a change, because a section cut for the wrong analysis cannot be uncut.

### ASK 12W

Cut to the priority on the board, and keep watching the board.

### BEHIND THE BUTTON 101W

Why the priority changes. Early the season is about the chronology, so the dating sections come first. When the gas laboratory's slot is confirmed the priority becomes anything for gas, which needs the core's centre and cannot use what has been near the wall. Before the aircraft it becomes anything that has to fly.

Why this is irreversible in a way most work is not. Every other mistake here can be repeated. A cut cannot: the section is gone, the depth it came from cannot be re-drilled, and what was cut for one analysis is unavailable to the others for ever.

### TAKES AS READ

a core section can only be cut up once

### VERDICT 136W

Three priorities run across the day and each wants a different part of the rack. Anything for the dating work, then anything for gas, then anything leaving on the aircraft. A section can answer two of those at once, which is what makes the change cost something real. What the panel scores is the window either side of each change, because most of the rack is wanted by neither priority and is correctly left alone by somebody who has read nothing. And this is the format's sharpest case, because a cut cannot be undone. Everything else on the plateau can be repeated with more time or more fuel; a section cut for isotopes is not available to the gas laboratory ever again, and the ice at that depth cannot be drilled a second time this season.

### TAKEAWAY 14W

The cost of a withdrawn instruction is highest where the work cannot be repeated.

## 9.5 How old the bottom of the firn is

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · CALLBACK · BALLPARK

### SCENE 31W

Brandt has the firn temperature string reading down to 88 metres, which is where the pores seal at this site. The surface is adding about 0.32 metres of snow a year.

### GUIDE 63W

Four numbers, and two of them belong to other questions: the ice-equivalent rate, and the density of solid ice. Ask of each whether it describes snow as it lies. In the firn the layers have not been thinned by flow, so this is a straight division. Deeper down the same sum has to be done interval by interval, because the layers get thinner.

### ASK 10W

Estimate the age of the snow at the close-off depth.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

a depth divided by an annual layer thickness gives a number of years · why ice keeps a record at all: burial without melting — taken as read · accumulation rate, and what a site with little snow buys and costs — taken as read

### VERDICT 89W

A depth–age scale is arithmetic on layer thicknesses: divide the depth by the thickness one year adds and you have the years it took to get there. In the firn the layers have not yet been thinned by flow, so the sum is a straight division; deeper in the ice the same calculation has to be done interval by interval as the layers get thinner. Nearly three centuries of snow sit above the close-off depth at Vestri, and every one of those years is still open to the atmosphere.

### TAKEAWAY 12W

A depth becomes an age only through the rate that filled it.

### END OF THE DAY 13W

The deepest ice holds the longest record and the weakest claim about it.

## Two records, side by side

### PLAN CARD 102W

Wednesday, and both records are on one screen for the first time. They differ by about half a per mil through the cold interval, which is larger than either group has claimed as its uncertainty. Halvorsen has a draft paragraph written. Okonkwo has asked what the two records have in common before anybody asks what they disagree about. Today you work out which parts of the comparison are independent, what the gas record says about the energy budget, and what half a per mil is worth in degrees. Two records that share a method are one record with two names on it.

### BRIEFING 14W

The Skarv record and this one differ by more than either group's stated uncertainty.

### WHAT YOU DECIDE 16W

Put both records on one plot and find out how much of the comparison is independent.

## 10.1 What both records rest on

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · TRACE

### SCENE 30W

Five pieces of the comparison are on Aalto's screen, each one computed from something. Okonkwo wants to know which of them would still stand if the flow model were wrong.

### GUIDE 53W

Open each piece of the comparison to see what it was computed from. Keep the ones that would still stand if the flow model were wrong, and untick the ones that run through it. Then name the shared dependency. A piece that survives is worth more than the number attached to it suggests.

### ASK 12W

Which parts of this comparison would survive the flow model being wrong?

### BEHIND THE BUTTON 143W

What a flow model does here. It converts depth into age by describing how the ice has thinned and moved since it fell. Almost every quantity that carries a date has passed through it, which is what makes it the most consequential assumption in the building.

Why some things survive it. Anything measured against a fixed marker — an ash layer, a known event, a directly counted section — is a claim about ice rather than about the model. Those hold whatever the model does, and they are how the model itself gets tested.

Why the question is asked this way round. Nobody expects the flow model to be wrong outright. The question is which conclusions are hostages to it, so that when it is revised — as it is every few years — the group knows exactly what has to be redone.

### TAKES AS READ

two results computed from one assumption fail together if that assumption fails

### VERDICT 90W

Both groups counted layers as far as they could and then handed the rest to the same flow model, with the same assumption of uniform vertical strain. Below those counts the two age scales are not independent estimates that happen to agree; they are one calculation applied twice. What survives the model being wrong is the counted section, which is arithmetic on measured layers, and the tephra horizon, whose date came from a volcano and not from any ice. Those two are the comparison. The rest is a shared assumption.

### TAKEAWAY 16W

Two numbers that agree because they came through the same step have not confirmed each other.

## 10.2 What the carbon dioxide change was worth

TRAPPED GAS · GAS LABORATORY · A PERSON · BALLPARK

### SCENE 30W

Adeyemi has carbon dioxide at 190 parts per million through the cold interval and 280 at the top of the core. Halvorsen wants the difference expressed as an energy imbalance.

### GUIDE 55W

Four numbers, and one of them is the isotope temperature slope, which belongs to another question. Ask of each whether this forcing uses it. And note the shape of the relationship. The bands are already partly saturated. Each extra molecule does less than the last, so the effect follows a logarithm rather than a proportion.

### ASK 8W

Estimate the radiative forcing between the two concentrations.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

a greenhouse gas reduces the rate at which the surface loses heat to space

### VERDICT 88W

Carbon dioxide absorbs in bands that are already partly saturated, so each extra molecule does less than the one before it, and the forcing goes as the logarithm of the concentration rather than in proportion to it. That is why the effect is usually stated per doubling: every doubling adds about the same. Between the cold interval and the top of the core the concentration rose by half, which is worth about two watts on every square metre of the planet, sustained for as long as it lasts.

### TAKEAWAY 12W

A concentration becomes a climate statement only through the energy it changes.

### 10.3 Half a per mil, in degrees

COLD LABORATORY · A ROOM · CHOICE

#### SCENE 31W

The two records differ by about half a per mil through the cold interval. Tanabe has the site calibration slope and Halvorsen's draft paragraph, which calls the difference a different climate.

#### GUIDE 66W

Four readings of half a per mil, and one of them refuses to convert at all. Ask of each whether it uses this site's own slope, and in which direction. Multiplying where you should divide gives an answer far too large. And note why the conversion matters. In per mil the difference sounds like rounding, and in degrees it is a quarter of the season's headline.

#### ASK 15W

What does half a per mil amount to, in the terms the draft paragraph uses?

#### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

#### TAKES AS READ

an isotope difference is converted to a temperature difference by the site slope

#### VERDICT 85W

The site slope is 0.67 per mil for each degree, so half a per mil is about 0.75 of a degree. That is the whole point of converting: in per mil the difference sounds like a rounding error, and in degrees it is a quarter of the cooling the season intends to report. Whether it is a real climate difference or a shared method's shortcoming is a separate question, and it cannot be argued about honestly until both sides are talking in the same units.

#### TAKEAWAY 17W

A difference has to be converted into the units of the claim before anybody can judge it.

## 10.4 The metre with no label

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CALLBACK · CHOICE

### SCENE 33W

A section has arrived at the line with a broken depth mark. Holt can place it to within about four metres from the run sheet, and the isotope team has asked for it.

### GUIDE 72W

Four options for a section that can only be placed within four metres. Ask of each what the depth is for. It is what turns a measurement into a statement about a year, and four metres here is thirty-odd years. Some work does not use depth at all. And note the real rule: the uncertainty has to travel with the sample, because the danger is somebody downstream who never hears about it.

### ASK 7W

What should the section be used for?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

a measurement belongs to the depth it came from, and to no other

### VERDICT 93W

A depth is what converts a measurement into a statement about a year. Four metres here is thirty-odd years, so an isotope value from this section cannot be placed in the record and would sit in it as a point with no time. It is still perfectly good ice for anything the depth does not enter: density, crystal fabric, how the drill is performing. The rule that matters is that the uncertainty travels with the sample, because the danger is not the gap — it is somebody downstream who never hears about it.

### TAKEAWAY 12W

A sample without a depth is a measurement of nothing in particular.

END OF THE DAY 11W

Agreement between two records is worth what their independence is worth.

## The offset that was not climate

### PLAN CARD 102W

Thursday, and Adeyemi has spent the night on the two groups' age scales. Skarv sits at four times this station's accumulation rate, which means its gas–ice age difference is a fraction of ours — and the comparison so far has been plotted with both gas records on their own ice timescales. Okonkwo has been saying since the first week that two clocks have to be matched first. Today you predict what the difference becomes once the correction is applied, apply it, and check it against the one date neither group calculated. Getting this wrong publishes a chronology error as a climate result.

### BRIEFING 14W

The two records may differ because one dates ice and the other dates gas.

### WHAT YOU DECIDE 14W

Test whether the age difference accounts for the disagreement before it is called one.

## 11.1 What the correction leaves behind

TRAPPED GAS · GAS LABORATORY · A ROOM · VERIFY

### SCENE 30W

Adeyemi has the age difference for both sites and the machinery to re-plot both gas records on their own gas timescales. The tephra horizon is dated independently of either calculation.

### GUIDE 54W

Lock what you expect the difference to become once both gas records sit on their own gas timescales. Then re-plot them and see. The tephra horizon is dated independently of either calculation, so it is the check that can tell you whether the correction did what you predicted rather than merely what you wanted.

### ASK 16W

What is left of the difference once both gas records are on their own gas timescales?

### BEHIND THE BUTTON 142W

What the correction does. Each site's gas is younger than the ice around it, by an amount that differs between sites. Putting each gas record on its own gas timescale removes that offset. Any difference remaining afterwards is a difference between the sites rather than between their chronologies.

Why a prediction first. If you re-plot and then decide what you expected, any residual difference can be explained after the fact. Committing to a number is what makes the result capable of surprising you, which is the only way it can be evidence.

Why the tephra matters. A volcanic ash layer arrives at both sites within days and is dated by neither calculation, so it is a fixed point both records must agree on. If the correction is right, the ash lines up. If it does not, something in the correction is wrong.

### TAKES AS READ

a correction should be predicted before it is applied, or it cannot be wrong

### VERDICT 90W

Air keeps mixing with the atmosphere until the firn seals, so gas is younger than the ice around it by an amount set by how deep the sealing is and how fast snow arrives. Skarv accumulates four times as fast, so its offset is a fraction of this station's.

Plotting both gas records on ice timescales therefore shifts one curve much further than the other, and produces a difference that looks like climate and is arithmetic.

What makes the corrected answer trustworthy is the tephra: a date neither calculation produced.

### TAKEAWAY 14W

A prediction made before the correction is the only thing that can test it.

## 11.2 The clock that does not need layers

CHRONOLOGY & THE CORE LINE · CORE LINE · A PERSON · BALLPARK

### SCENE 33W

Okonkwo has a sample from the deep section and a radiometric result: the isotope in it has fallen to 0.55 of the amount it starts with. Its mean lifetime is on the bench.

### GUIDE 65W

Four numbers, and two of them are traps. One is the half-life of the same isotope, which is not the same as its average life. One is the ice thickness. Ask of each whether this clock depends on it. And note why the answer is worth having. The decay rate cares nothing about temperature, pressure or flow, so it owes nothing to the flow model.

### ASK 13W

Estimate the age of the sample from what is left of the isotope.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

a radioactive isotope decays at a rate nothing in the ice can change

### VERDICT 92W

Radioactive decay proceeds at a rate that no temperature, pressure or flow in the ice can change, which is what makes it usable where the layers have stopped being countable. The amount left falls exponentially with time, so taking the natural logarithm of the fraction remaining and multiplying by the mean lifetime gives the age directly. It carries its own uncertainties — how much was there to begin with, whether anything was added or lost — and none of them is the flow model, which is why the result is worth having.

### TAKEAWAY 18W

Where counting has run out, a clock that runs on its own is the only independent date left.

## 11.3 What is left to disagree about

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · CHOICE

### SCENE 25W

Both records are re-plotted, the tephra horizon lines up in both, and 0.15 per mil of difference remains. Both groups quote uncertainties larger than that.

### GUIDE 64W

Four readings of what is left after the correction. Ask of each whether the remaining difference is bigger or smaller than the uncertainties either group quotes. And ask what a correction does to a result. This one was predicted before it was applied, then checked against a date that came from neither calculation. Both records were on the wrong clock; only one moved far.

### ASK 6W

What does the corrected comparison support?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

a difference smaller than the uncertainty of either measurement is not a measured difference

### VERDICT 91W

What remains after the correction is smaller than either group's stated uncertainty, so there is no measured difference left to explain. That is a positive result rather than an absence of one: two independently drilled cores, once put on comparable clocks, record the same cooling. The correction does not weaken it — it was predicted before it was applied and then checked against a date that came from neither calculation. And nothing here makes Skarv wrong; both records were plotted on the wrong clock, and only one of them moved far.

### TAKEAWAY 17W

Explaining a disagreement away is a result, and it is the result that has to be published.

## 11.4 Which year marker survives the depth

COLD LABORATORY · A ROOM · CALLBACK · CHOICE

### SCENE 27W

The isotope chemist, Yuki Tanabe, has 4 annual signals measured on the same stretch of core at 1,400 metres, where a year is about 26 millimetres thick.

### GUIDE 61W

Four annual signals, and a year here is twenty-six millimetres thick. Ask of each whether the thing being measured can move through the firn. Water molecules travel through open pores for decades, and that smooths anything thinner than about twice the distance they cover. Sea salt and mineral grains do not travel. And two of these fade for other reasons entirely.

### ASK 15W

At 1,400 metres, which signal is still able to separate 1 year from the next?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

water molecules can move through the connected air spaces in firn · firn: snow densifying into ice, and the close-off depth — taken as read · stable isotopes as a thermometer:  $\delta^{18}\text{O}$  and deuterium — taken as read

### VERDICT 84W

Water molecules move through the open pores of the firn for decades. That movement smooths the isotope record. Anything thinner than about twice the diffusion length is averaged away, and at this site that removes the yearly signal long before 1,400 metres. Sodium arrives as sea salt. Dust arrives as mineral grains. Neither travels through the pore space, so their yearly structure survives as far down as the measurement can read it. Visible layering fades once the bubbles are gone, and conductivity marks eruptions.

**TAKEAWAY 17W**

A signal is only useful for counting while it is still sharper than the layer it marks.

**END OF THE DAY 17W**

A disagreement between two records is a claim about both of them, including how they were dated.

## A day when nothing is wrong

### BEFORE THE DAY — EVADE

#### The film crew who want you in shot

There is a documentary team on the plateau this week and they want somebody standing at the drill looking thoughtful. The core comes up whether or not anybody is filming it, and a section that sits at the wrong temperature while somebody adjusts a light is a section nobody can use. Keep out of shot.

### PLAN CARD 108W

Friday, and for the first time this season nothing is broken. The drill is running clean, the chemistry line is caught up, and the camp manager, Anneke de Vries, needs the remaining plan before the fuel figures are committed. Nine drilling days are left and four jobs want them. Today you commit the drill time, settle what a single stake reading is worth, and fix the order a measurement run is made in. Nothing on the station is on fire, which makes this the only day of the season when the decision can be made on what is worth doing rather than on what has just gone wrong.

### BRIEFING 12W

The drill is running clean and there are nine drilling days left.

### WHAT YOU DECIDE 16W

Spend what is left of the season on the work that will still matter next year.

## 12.1 Nine days, four jobs

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · ALLOCATE

### SCENE 28W

Lindqvist has nine drilling days before the station closes. De Vries has the fuel committed for those nine and no more, and four pieces of work want them.

### GUIDE 57W

Nine drilling days, fuel committed for exactly nine, and four jobs that want them. Watch the questions list rather than the depths: it shows which questions the current plan can still answer. Every day you spend is a question you can answer and one you have given up, and nothing can be added after the station closes.

### ASK 9W

Which questions can the last nine days still answer?

### BEHIND THE BUTTON 140W

Why the fuel is the real constraint. Days can in principle be found; the fuel for those days cannot. It was committed before the season and there is no more, so the plan has to fit the fuel rather than the calendar.

Why depth is not the same as value. Drilling deeper always produces more core, and the extra core answers a question only if it reaches something the record needs. A shallower job that closes a comparison can be worth more than a deeper one that extends a series nobody is arguing about.

What the closing date does to the choice. Anything unfinished is unfinished for a year, and a core left mid-hole can be lost to closure or to the hole itself. So a plan that just fits is worse than one that fits with something in reserve.

### TAKES AS READ

a season ends on a date fixed by aircraft rather than by science

### VERDICT 97W

The two required questions are the ones this season exists to answer: whether the deep section reaches older ice, and whether the horizon everything is now tied to appears in ice that shared the weather but not the drill. Together with the fluid change they use exactly the nine days available. The firn profile and the borehole log are both good work and both can be done next season by somebody who has not yet drilled to 2,400 metres. A plan that fits everything in is a plan that has not been asked what it is for.

### TAKEAWAY 18W

A plan is judged by which questions it can still answer, not by how much it fits in.

## 12.2 The stake that gained a centimetre

SURFACE & ACCUMULATION · SNOW STUDY HUT · A PERSON · CHOICE

### SCENE 29W

Brandt has 48 stakes read this morning. One near the hut has gained a centimetre since the last reading. 11 others across the array have lost about the same.

### GUIDE 65W

Four readings of one stake that gained a centimetre while eleven others lost. Ask of each whether a single stake can separate snow that fell from snow that arrived sideways. Wind moves far more snow across this plateau in a fortnight than falls out of the sky. Summed across the array, the moving snow cancels. That is why forty-eight stakes are read rather than one.

### ASK 9W

What does the gain at that one stake establish?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

wind moves snow across a surface after it has fallen

### VERDICT 81W

Wind on this plateau moves far more snow horizontally than falls out of the sky in a fortnight, so any single stake is measuring snowfall and redistribution added together and cannot separate them. The array can: snow moved from one place to another cancels when the whole array is summed, and what is left is what actually arrived. That is why 48 stakes are read rather than 1, and why the accumulation figure is a season's arithmetic rather than a reading.

### TAKEAWAY 18W

A measurement of a surface that moves has to be made in enough places to catch what moved.

## 12.3 The order a run is made in

COLD LABORATORY · A ROOM · SEQUENCE

### SCENE 29W

Cruz has 16 samples to run and the instrument warmed up. Tanabe's rule is that a run has to be able to say afterwards whether it can be trusted.

### GUIDE 61W

All five of these will be done, so ask what each one makes possible. A blank says what the process adds with no ice in it. An opening standard fixes the scale. Standards inside the run reveal drift while it happens. Ask which of them a correction could be computed without. And note why both raw and corrected values are kept.

### ASK 6W

Fix the order of a run

### BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

### TAKES AS READ

an instrument's response can move slowly during a long run

### VERDICT 80W

Every step here exists to make the run auditable. The blank establishes what the process adds before any ice is involved; the opening standard fixes the scale; the standards inside the run reveal drift while it is happening; the closing standard measures how far it moved overall. Only then can the samples be corrected, and both the raw and the corrected values are kept because a correction is a claim about the instrument that somebody may want to check later.

### TAKEAWAY 17W

A run is designed so that it can be checked afterwards, not so that it finishes soonest.

## 12.4 The depth the counting runs out

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CALLBACK · SWEEP

### SCENE 55W

Snow buries snow, and each year's layer is squeezed thinner by the weight above it and the ice spreading below it. The scanner can separate two bands three millimetres apart and no closer. Okonkwo needs the depth where the annual layers thin to that limit, because past it the years cannot be counted at all.

### GUIDE 59W

Drag the depth. The readout is the measured thickness of one year's layer there, in millimetres, and only the depths you stop at get plotted. You are not looking for a peak: you want the depth at which the thickness falls to the scanner's three-millimetre limit. Sweep down the core, find that crossing, then mark the depth and commit.

### ASK 22W

The line resolves bands three millimetres apart and no closer. At what depth does the annual layer thickness fall to that limit?

### BEHIND THE BUTTON 158W

Why layers thin with depth. Ice under ice is compressed, and an ice sheet also spreads sideways under its own weight, stretching each buried layer thinner as it goes. Near the surface a year is centimetres of ice; two kilometres down the same year's snowfall may be a few millimetres.

What the limit costs. Above the crossing, a year is a band somebody can point at, and the age of the ice is a count. Below it, two years arrive inside one scanner resolution and the count stops. The ages then come from a flow model instead, which is an inference with its own uncertainty rather than a tally.

Why the crossing is a depth rather than a year. The thinning depends on the ice's own flow, so the same limit lands at different depths in different cores. It has to be measured in each one, which is why this reading is taken before any chronology is published.

### TAKES AS READ

layers are stretched thinner as the ice below them spreads outwards · accumulation rate, and what a site with little snow buys and costs — taken as read

### VERDICT 95W

Ice spreads outwards under its own weight, so every layer is stretched thinner as it sinks, and the thinning compounds with depth. At Vestri the surface lays down about 12 centimetres a year in ice equivalent; 2 kilometres down the same year is a few millimetres. Where the layers close to within the resolution of the measurement, the count stops — not because the years are gone, but because two of them can no longer be told apart. Everything deeper is dated by a flow model anchored to whatever dated events can still be found.

### TAKEAWAY 13W

A count ends where the instrument stops separating one year from the next.

#### END OF THE DAY 14W

The quiet days are when the season decides what it will have been about.

## The night the generator stopped

### PLAN CARD 106W

Saturday, and the number two generator stopped at some point in the night. The core store came back at minus 19 and the log has an entry for the whole night signed by the watch. De Vries needs the drill and the store back in a fixed order and Lindqvist wants his own list first. Aalto has been asked how much of the season's temperature record survives the scatter in it. Today you decide which signatures are worth checking, what gets restarted first, and how tightly a mean can be quoted. Ice that warmed and was signed for as cold is worse than ice that warmed.

### BRIEFING 15W

The store was without power for six hours and the log says nothing about it.

### WHAT YOU DECIDE 14W

Work out what the core store record can still prove after a power failure.

## 13.1 Six signatures and two hours

COLD LABORATORY · A ROOM · ATTEST

### SCENE 24W

Six lines on the night sheet, all signed. Tanabe has two hours before the flight manifest closes and can verify two of them properly.

### GUIDE 57W

Six signed lines, two hours, and two of them can be verified properly. Open each claim and read what already backs it. A signature records that somebody took responsibility, not that anybody observed the thing. Hold what the backing does not support, and spend the two hours where being wrong would matter after the flight has gone.

### ASK 8W

Which two claims should the two hours verify?

### BEHIND THE BUTTON 142W

Why the manifest deadline is the whole pressure. Once the flight closes, anything unverified stays unverified until next season, and the samples or the equipment it concerns are already off the ice. A claim checked next year is a claim about something nobody can revisit.

What a night sheet is. A running record of what was done on shift, signed at the end. It is written by people who were tired, at the end of a working night, and its purpose is continuity rather than evidence — which is exactly why some lines on it are firmer than others.

How to choose between two doubtful lines. Not by which is least likely to be true. By which one, if wrong, would change what somebody does with the material afterwards. That is consequence, and it is the judgement the two hours are for.

### TAKES AS READ

a signature records that somebody believed something, at the time they signed · random against systematic uncertainty — taken as read · separating a trend from year-to-year noise — taken as read

### VERDICT 95W

A signature says that somebody believed something when they signed, and the store reading of minus 19 says the belief was wrong for at least part of the night. Two of these claims are both critical and carried by nothing but the signature — the store and the gas samples — and both decide whether measurements made on that ice can be defended. The mast is critical and already carries its own continuous record; the drill tank has an independent logger. The food and the fuel matter to the camp and not to the science.

### TAKEAWAY 18W

A signed record is evidence that a claim was made, which is not evidence that it was true.

## 13.2 What comes back first

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · TRIAGE

### SCENE 30W

Power is back on one generator. Lindqvist, de Vries and Tanabe each want their own system restarted first, and the single generator will not carry all of them at once.

### GUIDE 56W

Four systems, and three of them are waiting. Ask of each whether the delay costs time or costs evidence. Ice above minus twenty starts losing gas from the bubbles near its cut faces, and nothing recovers it. A drilling day is a day. Queued samples are as good tomorrow. Order by what is still being lost.

### ASK 4W

What is restarted first?

### BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

### TAKES AS READ

a system that has already failed is more informative than one that has not yet been used

### VERDICT 81W

Three of these are waiting and one is deteriorating. Ice above minus 20 starts losing gas from the bubbles near its cut faces, and that loss cannot be undone or measured around; every hour is more of the core that can no longer answer a gas question. A lost drilling day is a day, the queued samples are as good tomorrow as today, and a report sent six hours late is a report sent. Order by what is still being lost.

### TAKEAWAY 17W

Restart in the order of what is still being lost, not in the order of who asked.

### 13.3 How much of it is noise

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · BALLPARK

#### SCENE 29W

Aalto has 30 years of reconstructed annual temperature from the top of the core. The year-to-year scatter is 1.4 degrees and the summary wants a mean for the period.

#### GUIDE 56W

Four numbers, and two of them belong to other questions: the calibration slope, and the years of mast overlap. Ask of each whether this uncertainty depends on it.

Averaging does not remove the scatter of individual years. It reduces the uncertainty of the mean, and only as the square root of how many years went in.

#### ASK 8W

Estimate the standard error of the thirty-year mean.

#### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

#### TAKES AS READ

averaging independent years reduces the scatter of the average, not of the years ·  
random against systematic uncertainty — taken as read

#### VERDICT 86W

Individual years scatter because weather scatters, and averaging does not remove that scatter — it reduces the uncertainty of the average, and only as the square root of the number of years. 30 years of a record with 1.4 degrees of year-to-year variation give a mean known to about a quarter of a degree. Two such means half a degree apart are two standard errors apart, which is where a claim of difference starts becoming defensible and not before. Doubling the precision would take 120 years.

#### TAKEAWAY 20W

An average is only as tight as the scatter divided by the root of how many years went into it.

## 13.4 A year, or a summer that melted

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CALLBACK · BELT

### SCENE 41W

Sections come out of the trench faster than they can be logged properly. A layer that formed over a year is one thing. A layer where the surface melted and refroze is another, and both are visible on the light table.

### GUIDE 60W

Two bins, and the question is whether the layer is a year of accumulation or a melt event inside one. An annual layer pairs a coarse summer with a fine winter and keeps its bubbles. A melt layer is clear, dense and bubble-poor, because the water filled the spaces before it refroze. Sort on the ice, not on the label.

### ASK 12W

Send each section to the bin that says what made the layer.

### BEHIND THE BUTTON 93W

Why the difference decides the chronology. The count of annual layers is the clock. A melt layer counted as a year makes the core older than it is at every depth below; a year missed makes it younger, and both errors propagate all the way down.

What melt does to everything else. Percolating water moves the chemistry down through the layers it passes, so a melt layer is not only a counting problem — it smears the record either side of it, which is why the site was chosen where melt is rare.

### TAKES AS READ

snow builds up in layers and can melt and refreeze

### VERDICT 171W

An annual layer is built by a season: coarse crystals from summer sun, fine wind-packed snow through winter, and air trapped as bubbles throughout. A melt layer is built by water. The surface thaws, the water percolates down and refreezes in the pore space, so what comes out on the light table is clear, dense and almost free of bubbles. Both are stripes in a core and only one of them is a year. The cost of confusing them is not local: annual layer counting is the clock, so a melt layer counted as a year makes every date below it too old, and a year missed makes every date too young, all the way to the bottom of the core. Melt also does something worse than miscount. Water moving down carries the chemistry with it, so the isotopes and the ions either side of the layer are smeared — which is why a site is chosen for how rarely its surface melts rather than for how much snow it gets.

### TAKEAWAY 15W

A layer counted wrong moves every date below it, in the same direction, for ever.

### END OF THE DAY 17W

A record is a claim about a condition, and the two can come apart in a night.

## What the season is allowed to say

### PLAN CARD 91W

Sunday, and the season report closes tomorrow morning. The age difference that explained most of the disagreement is itself a calculation with a range, and Halvorsen has four possible sentences on the board. Okonkwo will sign the one that survives the whole range and no other. Adeyemi has been asked what the gas record on its own establishes. Today you choose the sentence, settle what the carbon dioxide measurements license, and fix what has to happen before anything is sent. Tomorrow the season stops being something anybody can add evidence to.

### BRIEFING 17W

The report goes tomorrow, and the correction it rests on has a range rather than a value.

### WHAT YOU DECIDE 15W

Choose the claim that survives the range of the correction, and get the report out.

## 14.1 Four sentences and a range

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · STRESS

### SCENE 31W

The correction removes somewhere between a tenth and four tenths of a per mil of the difference, depending on assumptions nobody can settle this season. Four sentences are on the board.

### GUIDE 61W

The correction removes somewhere between a tenth and four tenths of a per mil, and nobody can settle which this season. Move it to the least favourable end before choosing between the four sentences. A sentence that is true at the middle of the range and false at the end of it is not a sentence the season summary can carry.

### ASK 15W

Which sentence would still be true at the least favourable end of the correction's range?

### BEHIND THE BUTTON 141W

Why the pessimistic end rather than the best estimate. A published sentence is read as a claim about the world, not about the middle of somebody's uncertainty.

Choosing at the expected value means publishing something that has a real chance of being wrong, and the retraction costs more than the caution would have.

What makes one sentence survive and another not. Sentences that assert a direction usually survive; sentences that assert a size usually do not, because the size is exactly what the correction's range moves. A carefully weakened claim can be both true and useful.

Why this cannot wait for the assumptions to settle. The summary is written this season and the assumptions will not be settled for years. The professional move is to write the strongest sentence that survives the whole range rather than the strongest sentence full stop.

### TAKES AS READ

a correction estimated from a model carries a range rather than a single value

### VERDICT 97W

The correction is calculated from a firm model, and firm models disagree with each other by more than anybody would like, so what comes out is a range. A sentence is publishable when it is still true at the least favourable end of that range. Saying the records are consistent needs the correction to do most of the work, and at the low end it does not. Saying the difference is entirely chronological needs more than the correction can do at any end. The hedged sentence is the slowest to write and the only one that survives.

### TAKEAWAY 20W

The claim to publish is the one that is still true at the least favourable end of what you assumed.

## 14.2 What the bubbles license

TRAPPED GAS · GAS LABORATORY · A PERSON · CHOICE

### SCENE 31W

Adeyemi has carbon dioxide through the cold interval and the modern section, measured on this core and reproduced on the replicate. Halvorsen wants to know what it supports on its own.

### GUIDE 65W

Four options, and the question is what this measurement is of. Ask of each how far the quantity travels before it is measured. Carbon dioxide stays in the air long enough to mix through all of it, so a bubble here reports the planet. The isotopes record condensation over this region, and reach degrees through a slope fitted at this site. Local at both steps.

### ASK 12W

What does the carbon dioxide record support that the isotope record cannot?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

a measurement made at one site can record a quantity that is the same everywhere

### VERDICT 75W

Carbon dioxide stays in the air long enough to mix through all of it before it is removed. Its concentration is therefore very nearly the same everywhere. A bubble from this plateau reports the whole planet's air, and it is the one global quantity this core measures directly. The isotopes are different. They record the condensation temperature over this region, and they reach degrees through a slope fitted at this site. Local at both steps.

### TAKEAWAY 19W

Some quantities measured at one place are global, and a proxy of local weather is not one of them.

### 14.3 Before anything is sent

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · SEQUENCE

#### SCENE 25W

Okonkwo has the sentence, the corrected records and a report that closes in the morning. Four things are outstanding and their order is not obvious.

#### GUIDE 66W

All five of these will be done before the report goes. Ask of each what the next step would mean without it. Nobody can reproduce a result without knowing which version of the age scales produced it. Recomputing from raw data tests that the draft's number is the method's number. And one of these turns a disagreement about another group's record into a conversation with them.

#### ASK 7W

Fix what happens before the report goes

#### BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

#### TAKES AS READ

a result should be reproducible by somebody who was not present

#### VERDICT 91W

Each step is what makes the next one mean something. Freezing the age scales fixes what the result was computed on, which is the only way anybody can reproduce it later. Recomputing from raw data on that frozen version tests that the number in the draft is the number the method produces. Writing the correction and its range into the methods is what lets a reader disagree with it specifically. Circulating to the other group is what turns a disagreement about their record into a conversation with them. Then it goes.

#### TAKEAWAY 16W

A result leaves the station as a package: the numbers, the version, and what was assumed.

## 14.4 The same water, twice

COLD LABORATORY · A ROOM · CALLBACK · CHOICE

### SCENE 31W

Tanabe has both groups' working standards on the bench and 30 minutes of instrument time. Skarv reports its values against its own standard, calibrated in a different laboratory four years ago.

### GUIDE 71W

Four things a joint run might establish. Ask of each whether one instrument in one afternoon could separate it from the alternatives. An isotope value is a difference from a reference water, so it carries whatever that reference is worth. Two laboratories keep their own. A scale gap and a real site difference both appear as a constant shift. That is why one of these needs both standards on one instrument.

### ASK 9W

What does running both standards on this instrument establish?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

an isotope value is reported as a difference from an agreed reference water

### VERDICT 91W

An isotope value is not an amount; it is a difference from a reference water, so it carries whatever that reference is worth. Two laboratories keep their own working standards, both traceable to the same international water and both slightly off it. Run side by side on one instrument, the two standards show the gap between the scales directly, in the same units the records are plotted in. Nothing else on the bench can separate that gap from a real difference between the sites, because both appear as a constant shift.

### TAKEAWAY 15W

A ratio reported against two different references is two different numbers about the same water.

**END OF THE DAY 14W**

A claim has to survive the worst credible value of everything it rests on.

## What is known, and how well

### BEFORE THE DAY — TAG

#### Catch Okonkwo before she goes into the counting room

Okonkwo, the chronology lead, has the layer count for the brittle section, and it is the only thing that puts today's depths on a year scale. Once she is in the counting room with the lights down she is not interrupted for anybody, and every date this afternoon rests on that count.

### PLAN CARD 105W

Monday, the last working day, and the aircraft comes on Wednesday. The report is sent and Halvorsen wants the season's own account of itself: what is established, what rests on a model, and what the next team has to do first. Brandt is closing the stake array and Tanabe has one question left about how fine the record can ever be. Today you settle which sentence the record supports, sort what the season established from what it assumed, and put a number on the shortest event this ice could still show. What the next team believes about this ice is what gets written down now.

### BRIEFING 17W

The last day is spent deciding what the record supports and what the next season must do.

### WHAT YOU DECIDE 14W

State what the season established, at what precision, and what it could not reach.

## 15.1 The sentence the record will carry

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · CHOICE

### SCENE 27W

Four versions of the season's headline claim, all of them true in some sense, on Aalto's screen. One of them is the one the evidence actually reaches.

### GUIDE 57W

Four sentences, all of them true in some sense. Ask of each what it claims beyond what was measured here. One calls two records independent, and below the counts they share a flow model. One rounds until the uncertainty disappears. One is a claim about every record at this latitude, and nobody here has read them all.

### ASK 7W

Which sentence does this season's evidence reach?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

a claim is supported when the evidence would have come out differently if it were false

### VERDICT 83W

The first sentence says what was measured, over what interval, and which part of it depends on something that was not measured here. The second calls the two records independent, which the shared flow model makes untrue below the counts. The third rounds the number until the uncertainty disappears into it and attributes agreement to a site whose comparison needed a correction. The fourth is a claim about every other record at this latitude, and nobody at this station has read them all.

### TAKEAWAY 13W

An honest claim states what was measured, over what period, with what uncertainty.

## 15.2 What rests on what

SURFACE & ACCUMULATION · SNOW STUDY HUT · A PERSON · PROTOCOL

### SCENE 23W

Brandt has the season's four headline numbers written on the hut wall and a column beside each one that nobody has filled in.

### GUIDE 63W

Four headline numbers, and four descriptions of how each was got. Pair them by asking how many steps sit between the instrument and the number. One is a direct surface measurement checkable next year. One passes through a slope fitted to eleven years. One is counted where counting worked and modelled below. And one is a solid measurement attached to a modelled date.

### ASK 4W

Sort established from assumed

### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

### TAKES AS READ

a number is only as strong as the weakest step between the ice and it

### VERDICT 91W

The four numbers are not the same kind of thing, and the report has to say so. The accumulation is a direct measurement of the surface and can be checked next year. The cooling is a measurement passed through a calibration, so it inherits the calibration's uncertainty. The age is counted where counting worked and modelled where it did not. The gas concentration is a solid measurement of a global quantity attached to a depth whose age is itself modelled — strong in what it measures, weaker in when it happened.

### TAKEAWAY 18W

Every number in a proxy record belongs to a class, and the class is part of the number.

### 15.3 The shortest thing this ice can show

COLD LABORATORY · A ROOM · BALLPARK

#### SCENE 32W

Tanabe has the firn diffusion length for this site and the surface layer thickness. The next team wants to know whether a single warm summer could ever be found in this core.

#### GUIDE 62W

Four numbers, and one of them is a year at two thousand four hundred metres rather than at the surface. Ask which layer thickness the question is about. Diffusion smooths anything thinner than about twice the distance the molecules travel. Divide that limit by a year's thickness and the answer is in years. At the surface this site can just hold one.

#### ASK 11W

Estimate the shortest event this record could still resolve, in years.

#### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

#### TAKES AS READ

water molecules move through firn before the pores seal, averaging nearby layers together

#### VERDICT 92W

Water molecules diffuse through the connected pores of the firn for as long as those pores are open, which averages each layer with its neighbours. The scale of that averaging is the diffusion length, and anything thinner than about twice it is smoothed away past recovery. Dividing that limit by the annual layer thickness gives the answer in years. At the surface this site can just about hold a single year; deeper, where the layers have thinned, the same diffusion length covers many years and the annual signal is gone for good.

#### TAKEAWAY 21W

A record has a shortest event it can show, and it is set by the site rather than by the instrument.

## 15.4 Which differences a shared instrument removes

TRAPPED GAS · GAS LABORATORY · A ROOM · CALLBACK · PROTOCOL

### SCENE 30W

The gas geochemist, Samuel Adeyemi, has four ways the two records could differ written on the board, and one working day in which both cores are in the same building.

### GUIDE 62W

Four ways the records could differ, and four kinds of evidence. Pair them by asking what sharing an instrument can and cannot remove. A stretched timescale is a chronology problem and needs an event dated in both cores. A real climate difference is what survives after everything else is eliminated. And melting alters the outside of a section more than the inside.

### ASK 9W

Sort what a joint run can and cannot settle

### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

### TAKES AS READ

two measurements of one quantity can differ for reasons that have nothing to do with the quantity · accumulation rate, and what a site with little snow buys and costs — taken as read · annual layer counting, and where counting fails — taken as read

### VERDICT 90W

Only the first of these is fixed by sharing an instrument. A stretched timescale is a chronology problem and needs an event of known date present in both cores. A real climate difference is what is left once every other explanation has been removed, which is why it is established by elimination rather than measured directly. Melting in transit alters the outer ice more than the inner, so it leaves a signature in the

profile across a section rather than in the average. Four differences, four different pieces of evidence.

**TAKEAWAY 13W**

Sharing an instrument removes the differences the instrument was causing, and only those.

**END OF THE DAY 11W**

A record is worth what its limits are known to be.

## THE CLOSING CARD

The season closed with the hole at 2,470 metres and a record that reaches, by counting, to about nine thousand nine hundred years, and by flow model to something under thirty thousand. The disagreement with Skarv turned out to be mostly the gas-ice age difference: four hundred and fifty-odd years at this site and a fraction of that at theirs, applied to both records and then checked against a tephra horizon that neither calculation produced. What was left of the difference was smaller than either group had claimed as its uncertainty, and the report said so in the hedged sentence rather than the confident one.

What it cost: a night of the core store above minus twenty, which took the gas measurements from four sections with it; a fortnight in which the comparison was argued about on two timescales that could not be compared; and the deep dating, which still rests on a flow model this station did not build. What is unfinished: below the counted section every age is modelled, the firn air profile was never re-measured, and the next team arrives to a question nobody here could answer — whether the same correction holds at four times the accumulation rate, which is the assumption the whole comparison now stands on.

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